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CDC 09 2022 DILR

Scorecard (procreview.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:47:35 IST 2023&qsetId=0b5kV1s2XBA=&qsetName=CDC 09 2022 DILR)

Accuracy (AccSelectGraph.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:47:35 IST 2023&qsetId=0b5kV1s2XBA=&qsetName=CDC 09 2022 DILR)

Qs Analysis (QsAnalysis.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:47:35 IST 2023&qsetId=0b5kV1s2XBA=&qsetName=CDC 09 2022 DILR)

Video Attempt / Solution (VideoAnalysis.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:47:35 IST 2023&qsetId=0b5kV1s2XBA=&qsetName=CDC 09 2022 DILR)

Solutions (Solution.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:47:35 IST 2023&qsetId=0b5kV1s2XBA=&qsetName=CDC 09 2022 DILR)

Bookmarks (Bookmarks.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:47:35 IST 2023&qsetId=0b5kV1s2XBA=&qsetName=CDC 09 2022 DILR)

Section-1

Sec 1

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

In a Cook-House workshop for 100 days, participants registered, face-off against each other for scholarship. The workshop allows participants a maximum of 100 days to master 5 main dishes wherein each dish takes at least 2 days to be mastered in any order of their choosing. No dish is mastered simultaneously. Each participant is scored on each dish on a scale of 0 to 20 to a total of maximum 100 points for the 5 dishes.
Further, the rules for the participants are:
(i) The score of each participant, in points, at any point during the workshop will be **Score Overall = Sum of scores on dishe(s) completed at that point + Bonuses (if any) – Penalties (If any) – Total number of days taken to complete dishe(s) till that point.**
(ii) If any participant masters his/her first chosen dish within 7 days, scores at least 15 points on that dish as well as he/she will get 9 bonus points. However, if any participant takes 19 days or more to master the first dish, 5 points will be deducted as penalty.
(iii) If a participant masters the first 2 dishes within 30 days, scores at least 16 points on an average per dish as well as he/she will get 12 bonus points. However, if the participant is unable to master the first two dishes by the end of day 47, 10 points will be deducted as penalty.
(iv) If a participant masters the first 3 dishes within 40 days, scores at least 12 points on an average per dish as well as he/she will get a staggering 27 bonus points, while a failure to do so within 70 days will see him/her lose 21 points as penalty.
(v) Finally, mastering 4 dishes within 71 days will see a participant earn 12 bonus points while not finishing even by day 91 will see them lose 20 points as penalty.

Q.1 [11831809]
Binu and Vinu mastered exactly 4 dishes. Binu mastered her first 2 dishes in 36 days and the next two in another 35 days, winning the bonus only once. Vinu mastered 4 dishes gaining bonus points on each dish. What could be absolute difference, in points, between Binu's maximum possible score and Vinu's minimum possible score for the first 4 dishes?

1 ◯ 4

2 ◯ 10

3 ◯ 16

4 ◯ More than 20 points

Solution:
Correct Answer : 1

Answer key/Solution

First, maximise Binu's score and ensure that she only won bonus points only once, by arranging her efforts in a way that the only bonus points she wins is the one of completing the first 4 dishes within 71 days (she finishes her 4th dish on the 71st day so that bonus is hers for sure).
Thus, the data can be arranged in the following manner:

1st dish - More than 7 days	Score - 20
2nd dish - More than 30 days	Score - 20
3rd dish - More than 40 days	Score - 20
4th dish - Masters on the 71st day	Score - 20

Binu's maximum tally = 20 × 4 (20 points/dish) + 12 – 71 (sum of days) = 21 points
To minimise Vinu's total score, give him minimum possible scores while ensuring he wins all bonuses available.
Thus, his efforts will look like the following:

1st dish - on 7th day	Score - 15 (to win bonus)
2nd dish - on 30th day	Score - 17 (to win bonus, average ≥ 16 for first two dishes)
3rd dish - on 40th day	Score - 4 (to win bonus, average ≥ 12 for first three dishes)
4th dish - Masters on the 71st day	Score - 0

Vinu's minimum tally = 36 (dishes) + 60 (from 4 bonuses) – 71 (sum of days) = 25 points
Hence, absolute difference = 25 – 21 = 4 points.

Bookmark FeedBack

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(v) Finally, mastering 4 dishes within 71 days will see a participant earn 12 bonus points while not finishing even by day 91 will see them lose 20 points as penalty.

Q.2 [11831809]
If Rinu scored a total of 20 points even after she mastered exactly two dishes in 28 days, which of the following COULD represent the number of days she took to master her second dish alone?

1 ◯ 25

2 ◯ 22

3 ◯ 20

4 ◯ 9

Solution:
Correct Answer : 3

Answer key/Solution

If Rinu finished 2 dishes in 28 days, and her score is 20, we can say:
SCORE = Sum of scores on Dishes ± Bonuses/ Penalties (if any) – No. of days
⇒ 20 = Sum of scores on dishes ± bonuses/ penalities – 28
⇒ 48 = Sum of scores on dishes ± bonuses/ penalties
It means Rinu has to get at least one bonus because her score exceeds 40.
Now, use the options to solve.
Option 1: 2nd dish in 25 days means 1st dish in 6 days. Rinu is eligible for both bonuses.
Case I: If she wins both bonuses,
Sum of scores on dishes = 48 – 9 – 12 = 27 (not possible; she must score 32 in her first two dishes to win a bonus after the 2nd dish)
Case II: If she wins one of the bonuses,
Sum of scores on dishes = 48 – 9 / 12 = 39/ 36 (not possible; these scores correspond to a scenario where she would need to score at least 16 points in the first dish. She would thus be eligible to win both bonuses.)
Both scenarios are not possible. So option 1 is not possible.
Option 2: will give the same result as option 1. Therefore, it is also not possible.
Option 3: 2nd dish in 20 days, means 1st dish in 8 days. So bonus received will be from the 2nd dish only.
So sum of scores on dishes = 48 – 12 = 36
This score is possible since she would get a bonus on scoring 36 points from the first two dishes.
Option 4: 2nd dish in 9 days, means 1st dish in 19 days. So bonus received will be from the 2nd dish only (12 points) but she will be penalised for taking 19 days to master her first dish (5 points).
So sum of scores on dishes + 12 – 5 = 48
Sum of scores on dishes = 41
This score is not possible. Since scoring 41 points from two dishes is not possible.
Hence, option (3) is the correct answer.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

In a Cook-House workshop for 100 days, participants registered, face-off against each other for scholarship. The workshop allows participants a maximum of 100 days to master 5 main dishes wherein each dish takes at least 2 days to be mastered in any order of their choosing. No dish is mastered simultaneously. Each participant is scored on each dish on a scale of 0 to 20 to a total of maximum 100 points for the 5 dishes.
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(v) Finally, mastering 4 dishes within 71 days will see a participant earn 12 bonus points while not finishing even by day 91 will see them lose 20 points as penalty.

Q.3 [11831809]
If Tina, received bonus points twice but did not lose any points as penalty, mastering exactly four dishes in 80 days, then among the given options what could be her minimum possible score?

1 ◯ –15

2 ◯ –18

3 ◯ –23

4 ◯ –27

Solution:
Correct Answer : 4

Answer key/Solution

To minimise Tina's score, we need to minimise her bonus points as well as scores on the dishes.
Since she has won bonus points two times, let those be for the first two dishes.
Hence, Tinu's minimum score = 32 points (minimum needed to win bonus from first two dishes) + 0 + 0 (no points from next two dishes) + 21 (bonus) – 80 (days) = – 27 points.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

In a Cook-House workshop for 100 days, participants registered, face-off against each other for scholarship. The workshop allows participants a maximum of 100 days to master 5 main dishes wherein each dish takes at least 2 days to be mastered in any order of their choosing. No dish is mastered simultaneously. Each participant is scored on each dish on a scale of 0 to 20 to a total of maximum 100 points for the 5 dishes.
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(v) Finally, mastering 4 dishes within 71 days will see a participant earn 12 bonus points while not finishing even by day 91 will see them lose 20 points as penalty.

Q.4 [11831809]
Parth on mastering his first three dishes scored 14, 18 and 19, in no particular order but did not win bonus points after any dish. Kara, meanwhile, won bonus points twice while getting penalised once, after mastering 3 dishes. The minimum possible value of the sum of days taken by Parth alone and Kara alone is

Solution:
Correct Answer : 64

Answer key/Solution

To ensure Parth does not win any bonus points, we can arrange his efforts on the first three dishes to ensure he masters the third dish on the 41st day (minimum possible).
Kara's efforts can be arranged in a way to ensure that that she is penalised on the first dish but masters the other two dishes in two days each. Number of days (minimum possible) will thus, be 23 days (19 + 2 + 2).
Hence, sum = 41 + 23 = 64 days.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Two hundred eighty (280) students in a HR-MBA class of XYZ college gets 3 assignments and 2 research papers to submit within three days – Friday, Saturday and Sunday. It was observed that a student either submitted the assignments or they submitted the research papers but not both.
Further, the following information is also known that:
(i) The numbers of students who submitted atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.
(ii) The numbers of students who submitted exactly one assignment each day is 39.
(iii) The number of students who submitted all assignments on Saturday is the same as the number of students who submitted all their assignments on Sunday. This number is 20% of the number of students who submitted all their assignments on Friday.
(iv) No student submitted any research paper on Sunday. Of all the students who submitted the research papers, only 10 students summitted the research papers on both days.
(v) The number of students who submitted atleast one research paper on Saturday is 60% of the students who submitted all the assignments on Friday.
(vi) The number of students who submitted at least one research paper on Friday and the number of students who submitted all 3 assignments in two days Saturday and Sunday only are in the ratio of 10 : 1.

The professor decides to award grace marks per submission to the students based on how early they submit the assignment or research paper.

	Friday	Saturday	Sunday
Grace marks per Assignment	10	5	1
Grace marks per Research paper	15	10	

The total grace marks professor awarded to the students who submitted all research papers either on Friday or Saturday is 1300.

Q.5 [11831809]
How many students submitted all the assignments together on any day?

1 ◯ 105

2 ◯ 92

3 ◯ 98

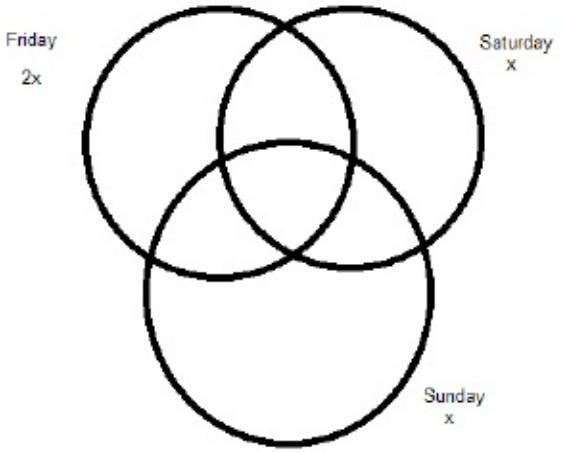
4 ◯ 112

Solution:
Correct Answer : 1

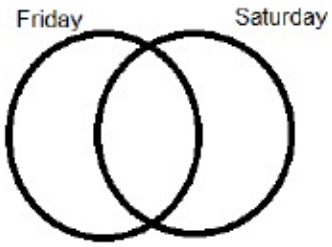
Answer key/Solution

Step 1:
From condition (i), The numbers of students who submitted the atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.

Assiginment

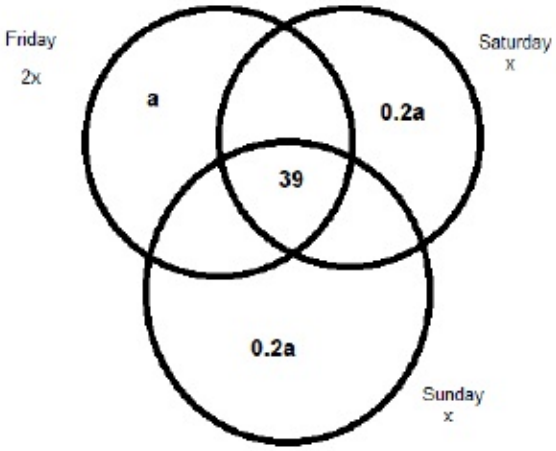


Research papers

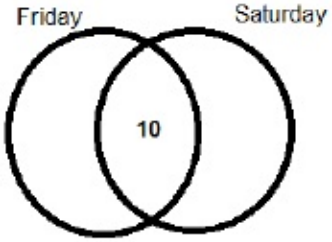


Step 2:
From conditions (ii) and (iii), The numbers of students who submitted assignment exactly one assignment every day is 39. Let number of students submitting all three assignments on Friday be 'a'.

Assiginment

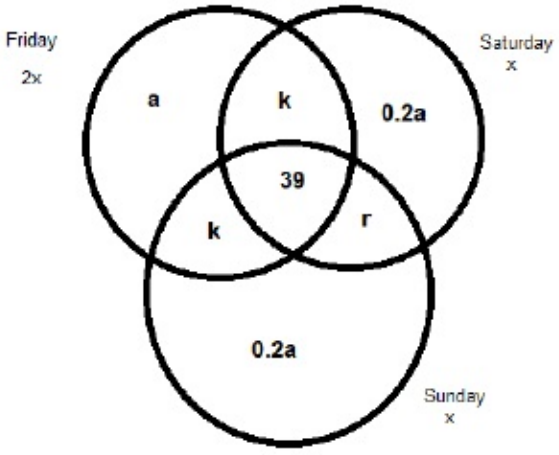


Research papers

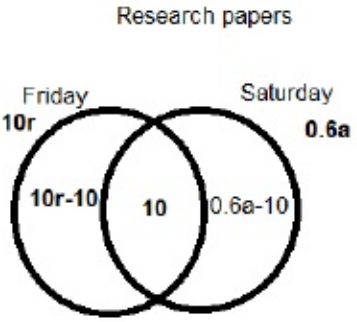


Step 3:
Also, from condition (iv), (v) and (vi), we get

Assiginment



Research papers



Number of students who submitted atleast one assignment on Saturday is equal to number of students who submitted exactly one assignment on Sunday. Therefore,

$M + 0.2a + 39 + r = N + 39 + r + 0.2a$

$\Rightarrow M = N = k.$

Step 4:

$1.4a + 2k + r + 39 + 10r - 10 + 10 + 0.6a - 10 = 280$

$\Rightarrow 2a + 2k + 11r = 251$ ... (1)
$a + 2k + 39 = 2(k + r + 39 + 0.2a)$
$\Rightarrow 0.6a - 39 = 2r$ ... (2)
$2(10r - 10) \times 15 + 2(0.6a - 10) \times 10 = 1300$
$\Rightarrow 150r + 6a = 900$ ... (3)
From (2) and (3), we get
r = 3, a = 75
Putting these values in (1), we get
$2a + 2k + 11r = 251$
$\Rightarrow k = 34$
Hence, final Venn diagrams will look like:

105 students submitted all the assignments together on any day.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Two hundred eighty (280) students in a HR-MBA class of XYZ college gets 3 assignments and 2 research papers to submit within three days – Friday, Saturday and Sunday. It was observed that a student either submitted the assignments or they submitted the research papers but not both.

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(v) The number of students who submitted atleast one research paper on Saturday is 60% of the students who submitted all the assignments on Friday.

(vi) The number of students who submitted at least one research paper on Friday and the number of students who submitted all 3 assignments in two days Saturday and Sunday only are in the ratio of 10 : 1.

The professor decides to award grace marks per submission to the students based on how early they submit the assignment or research paper.

	Friday	Saturday	Sunday
Grace marks per Assignment	10	5	1
Grace marks per Research paper	15	10	

The total grace marks professor awarded to the students who submitted all research papers either on Friday or Saturday is 1300.

Q.6 [11831809]

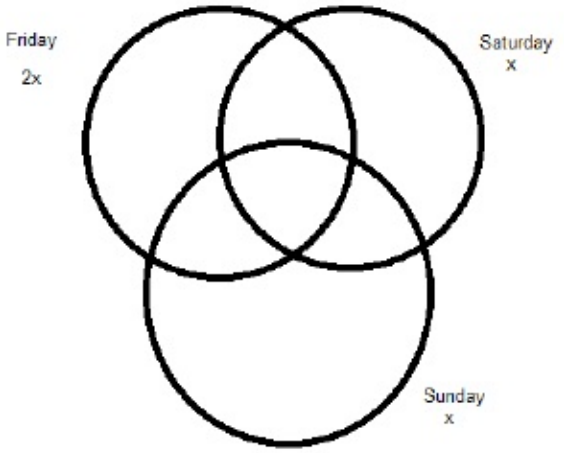
How many students submitted atleast one assignment on Sunday?

Solution:
Correct Answer : 91

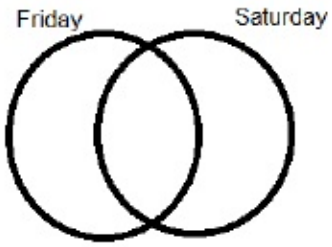
Answer key/Solution

Step 1:
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Assiginment

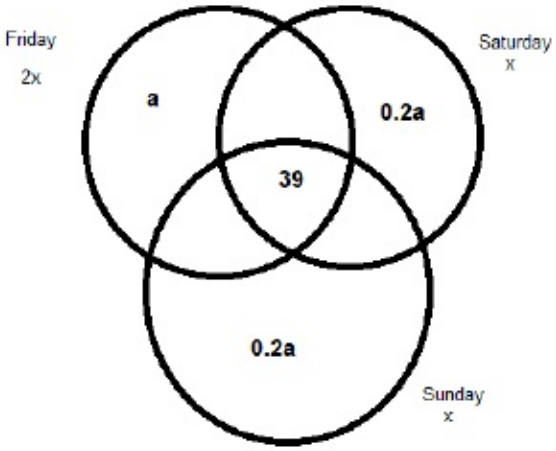


Research papers

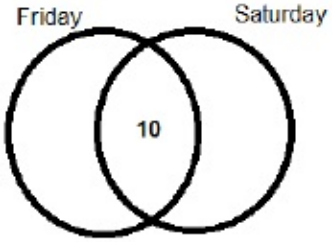


Step 2:
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Assiginment

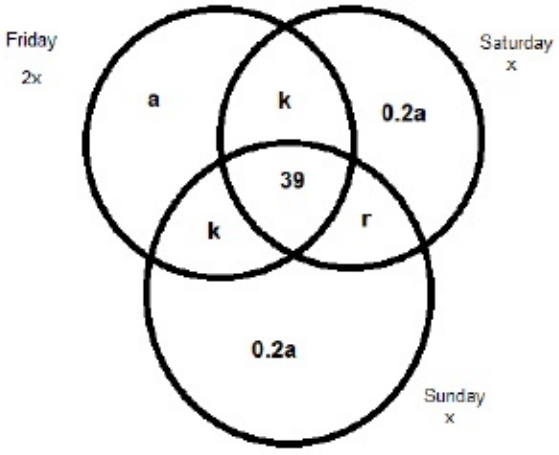


Research papers

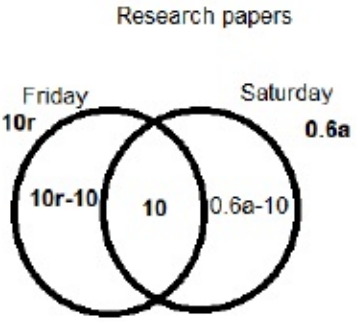


Step 3:
Also, from condition (iv), (v) and (vi), we get

Assiginment



Research papers



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$M + 0.2a + 39 + r = N + 39 + r + 0.2a$

$\Rightarrow M = N = k.$

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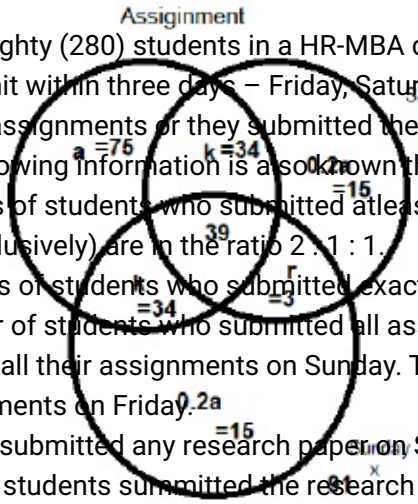
$1.4a + 2k + r + 39 + 10r - 10 + 10 + 0.6a - 10 = 280$

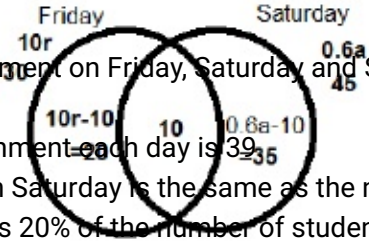
$\Rightarrow 2a + 2k + 11r = 251$ ... (1)
$a + 2k + 39 = 2(k + r + 39 + 0.2a)$
$\Rightarrow 0.6a - 3r = 21$ ... (2)
$2(10r - 10) \times 15 + 2(0.6a - 10) \times 10 = 1300$
$\Rightarrow 150r + 6a = 900$ ... (3)
From (2) and (3), we get
r = 3, a = 75
Putting these values in (1), we get
$2a + 2k + 11r = 251$
$\Rightarrow k = 34$
Hence, final Venn diagrams will look like:

91 students submitted atleast one assignment on Sunday.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.





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Grace marks per Research paper	15	10	

The total grace marks professor awarded to the students who submitted all research papers either on Friday or Saturday is 1300.

Q.7 [11831809]
The grace marks awarded by the professor to the students who submitted either one or two assignments on Saturday and Sunday lies between:

1 ◯ 500 and 700

2 ◯ 400 and 600

3 ◯ 400 and 700

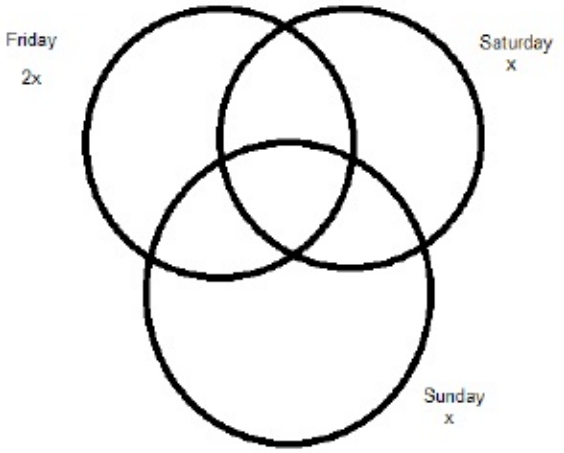
4 ◯ 500 and 800

Solution:
Correct Answer : 3

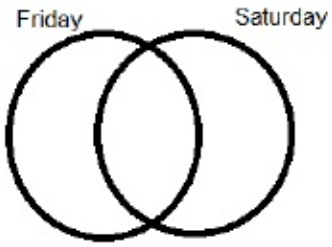
Answer key/Solution

Step 1:
From condition (i), The numbers of students who submitted the atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.

Assiginment

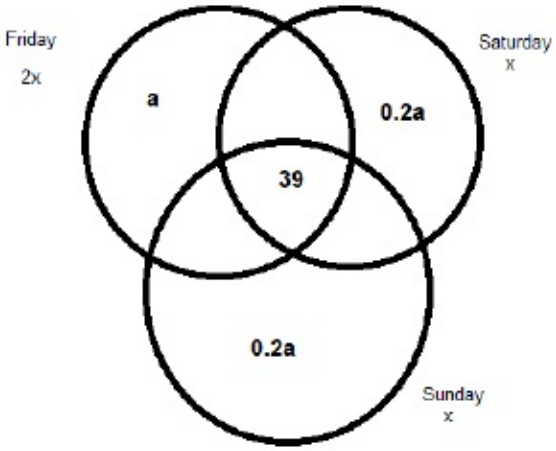


Research papers

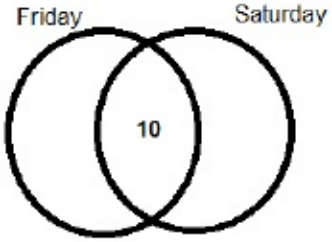


Step 2:
From conditions (ii) and (iii), The numbers of students who submitted assignment exactly one assignment every day is 39. Let number of students submitting all three assignments on Friday be 'a'.

Assiginment

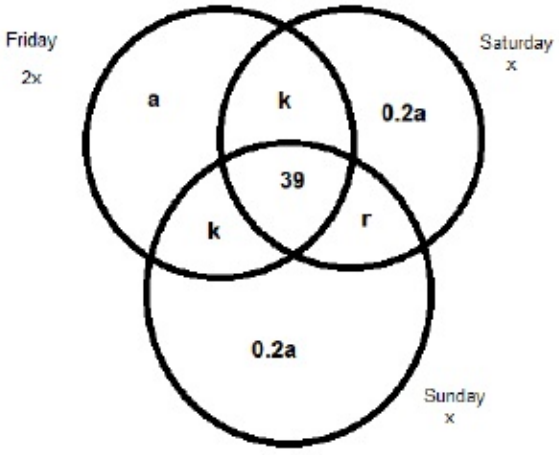


Research papers

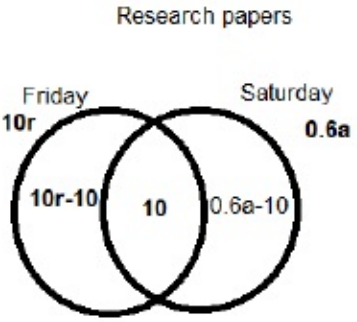


Step 3:
Also, from condition (iv), (v) and (vi), we get

Assiginment



Research papers



Number of students who submitted atleast one assignment on Saturday is equal to number of students who submitted exactly one assignment on Sunday. Therefore,

$M + 0.2a + 39 + r = N + 39 + r + 0.2a$

$\Rightarrow M = N = k.$

Step 4:

$1.4a + 2k + r + 39 + 10r - 10 + 10 + 0.6a - 10 = 280$

To find the interval, determine the minimum and maximum possible total grace marks provided to students by the professors.

For minimum possible:

Students who submitted all three assignments in two days Friday and Saturday are 34. Therefore, students must have submitted either 1 or 2 assignments on Saturday. To minimize consider that student submitted exactly 1 assignment on Saturday. Grace marks they got will be 34 × 5 = 170.

Similarly, students who submitted all three assignments in two days Friday and Sunday are 34. Therefore, students must have submitted either 1 or 2 assignments on Sunday. To minimize consider that student submitted exactly 1 assignment on Sunday. Grace marks they got will be 34 × 1 = 34.

3 students submitted assignments in two days Saturday and Sunday. To minimize let 2 assignments be submitted on Sunday and 1 on Saturday. Total Grace marks = 3 × 5 + 3 × 2 × 1 = 21

39 students submitted exactly 1 assignment every day. Total grace marks = 39 × 5 + 39 × 1 = 234.

Minimum possible total Grace marks = 459.

Similarly, maximum possible total grace marks will be = 675.

Hence, the possible total grace marks will be lie between 400 and 700

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Two hundred eighty (280) students in a HR-MBA class of XYZ college gets 3 assignments and 2 research papers to submit within three days – Friday, Saturday and Sunday. It was observed that a student either submitted the assignments or they submitted the research papers but not both.
Further, the following information is also known that:
(i) The numbers of students who submitted atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.
(ii) The numbers of students who submitted exactly one assignment each day is 39.
(iii) The number of students who submitted all assignments on Saturday is the same as the number of students who submitted all their assignments on Sunday. This number is 20% of the number of students who submitted all their assignments on Friday.
(iv) No student submitted any research paper on Sunday. Of all the students who submitted the research papers, only 10 students summitted the research papers on both days.
(v) The number of students who submitted atleast one research paper on Saturday is 60% of the students who submitted all the assignments on Friday.
(vi) The number of students who submitted at least one research paper on Friday and the number of students who submitted all 3 assignments in two days Saturday and Sunday only are in the ratio of 10 : 1.

The professor decides to award grace marks per submission to the students based on how early they submit the assignment or research paper.

	Friday	Saturday	Sunday
Grace marks per Assignment	10	5	1
Grace marks per Research paper	15	10	

The total grace marks professor awarded to the students who submitted all research papers either on Friday or Saturday is 1300.

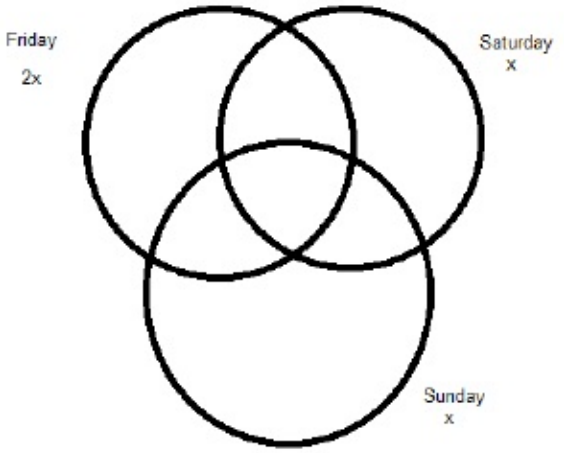
Q.8 [11831809]
How many students submitted atleast one research paper on Friday?

Solution:
Correct Answer : 30

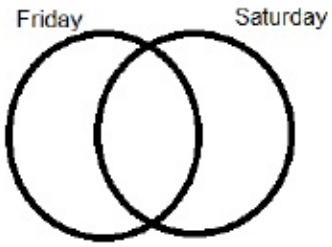
Answer key/Solution

Step 1:
From condition (i), The numbers of students who submitted the atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.

Assiginment

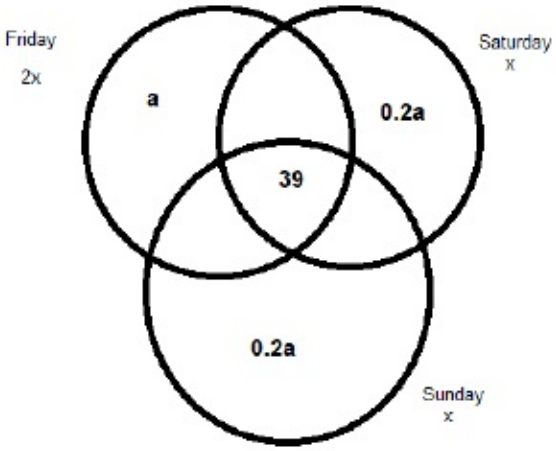


Research papers

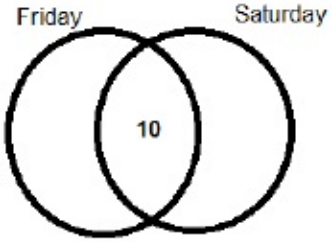


Step 2:
From conditions (ii) and (iii), The numbers of students who submitted assignment exactly one assignment every day is 39. Let number of students submitting all three assignments on Friday be 'a'.

Assiginment

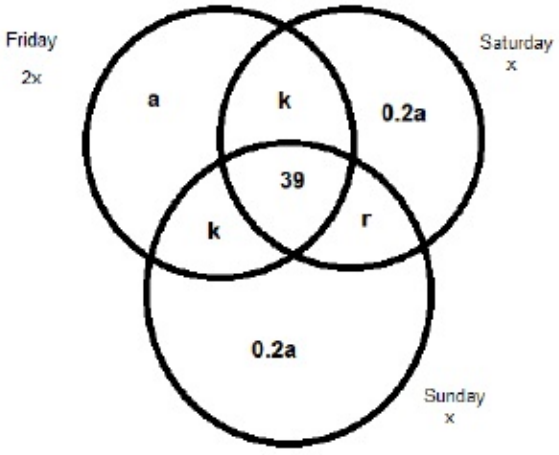


Research papers

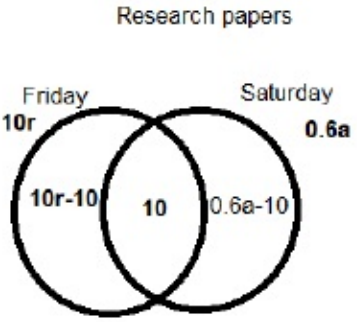


Step 3:
Also, from condition (iv), (v) and (vi), we get

Assiginment



Research papers



Number of students who submitted atleast one assignment on Saturday is equal to number of students who submitted exactly one assignment on Sunday. Therefore,

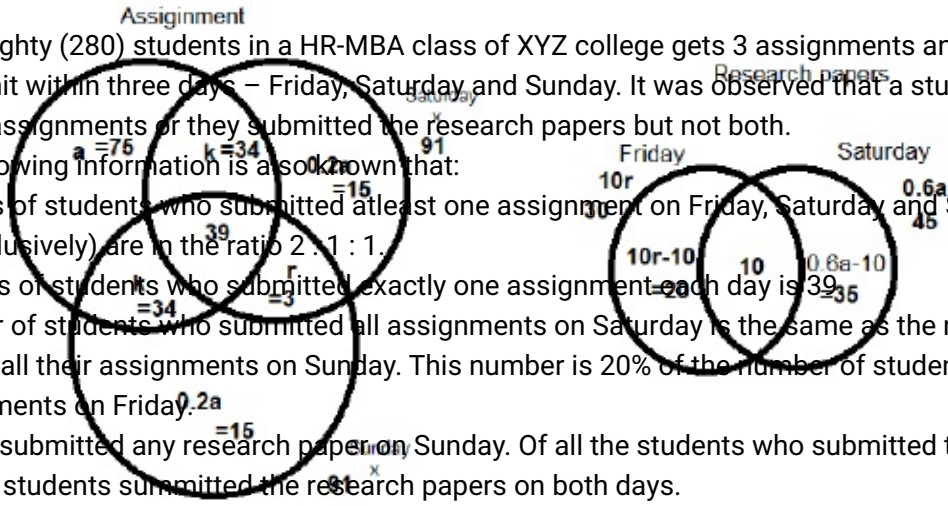
$M + 0.2a + 39 + r = N + 39 + r + 0.2a$

$\Rightarrow M = N = k.$

Step 4:

$1.4a + 2k + r + 39 + 10r - 10 + 10 + 0.6a - 10 = 280$

$\Rightarrow 2a + 2k + 11r = 251$ ... (1)
$a + 2k + 39 = 2(k + r + 39 + 0.2a)$
$\Rightarrow 0.6a - 39 = 2r$ ... (2)
$2(10r - 10) \times 15 + 2(0.6a - 10) \times 10 = 1300$
$\Rightarrow 150r + 6a = 900$ ... (3)
From (2) and (3), we get
r = 3, a = 75
Putting these values in (1), we get
$2a + 2k + 11r = 251$
$\Rightarrow k = 34$
Hence, final Venn diagrams will look like:



30 students submitted atleast one research paper on Friday.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Two hundred eighty (280) students in a HR-MBA class of XYZ college gets 3 assignments and 2 research papers to submit within three days – Friday, Saturday and Sunday. It was observed that a student either submitted the assignments or they submitted the research papers but not both.
Further, the following information is also known that:
(i) The numbers of students who submitted atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.
(ii) The numbers of students who submitted exactly one assignment each day is 39.
(iii) The number of students who submitted all assignments on Saturday is the same as the number of students who submitted all their assignments on Sunday. This number is 20% of the number of students who submitted all their assignments on Friday.
(iv) No student submitted any research paper on Sunday. Of all the students who submitted the research papers, only 10 students submitted the research papers on both days.
(v) The number of students who submitted atleast one research paper on Saturday is 60% of the students who submitted all the assignments on Friday.
(vi) The number of students who submitted at least one research paper on Friday and the number of students who submitted all 3 assignments in two days Saturday and Sunday only are in the ratio of 10 : 1.

The professor decides to award grace marks per submission to the students based on how early they submit the assignment or research paper.

	Friday	Saturday	Sunday
Grace marks per Assignment	10	5	1
Grace marks per Research paper	15	10	

The total grace marks professor awarded to the students who submitted all research papers either on Friday or Saturday is 1300.

Q.9 [11831809]
What can be said about the grace marks awarded by the professor in the research papers submitted on Saturday?

1 ◯ Must be between 400 and 800

2 ◯ Cannot be more than 800

3 ◯ More than 800

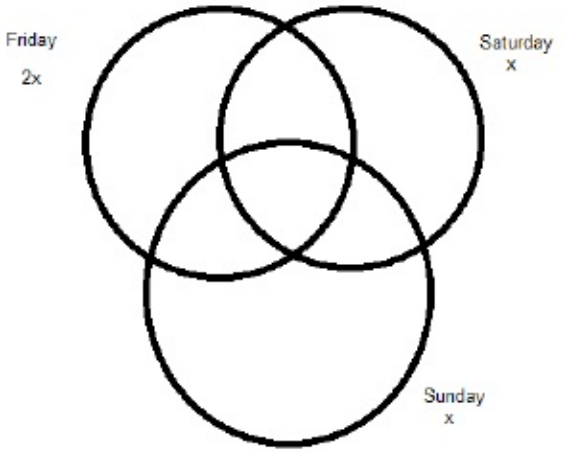
4 ◯ Must be between 650 and 750

Solution:
Correct Answer : 2

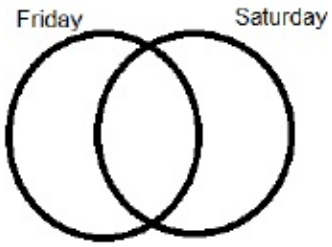
Answer key/Solution

Step 1:
From condition (i), The numbers of students who submitted the atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.

Assiginment

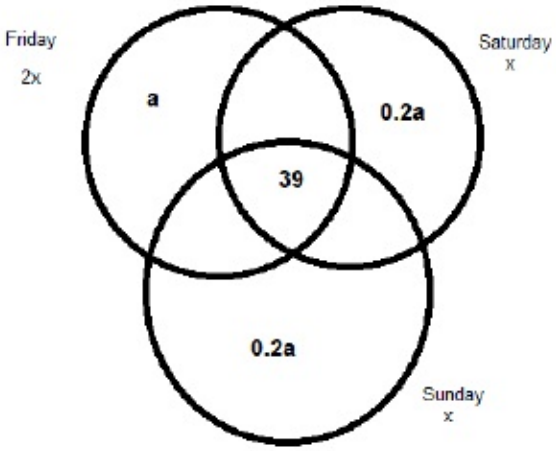


Research papers

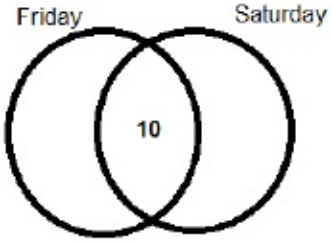


Step 2:
From conditions (ii) and (iii), The numbers of students who submitted assignment exactly one assignment every day is 39. Let number of students submitting all three assignments on Friday be 'a'.

Assiginment

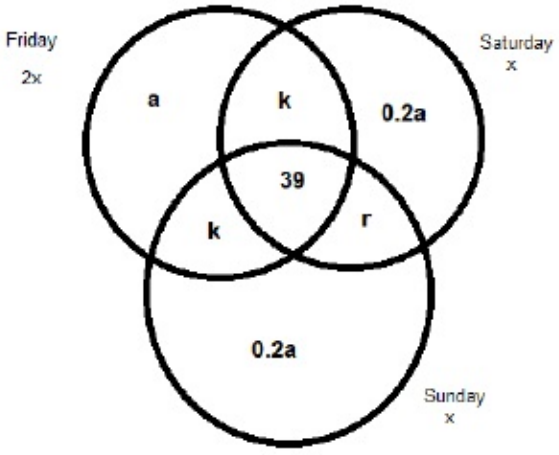


Research papers

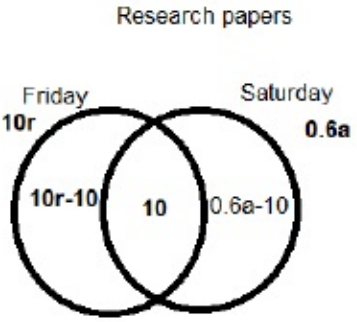


Step 3:
Also, from condition (iv), (v) and (vi), we get

Assiginment



Research papers



Number of students who submitted atleast one assignment on Saturday is equal to number of students who submitted exactly one assignment on Sunday. Therefore,

$M + 0.2a + 39 + r = N + 39 + r + 0.2a$

$\Rightarrow M = N = k.$

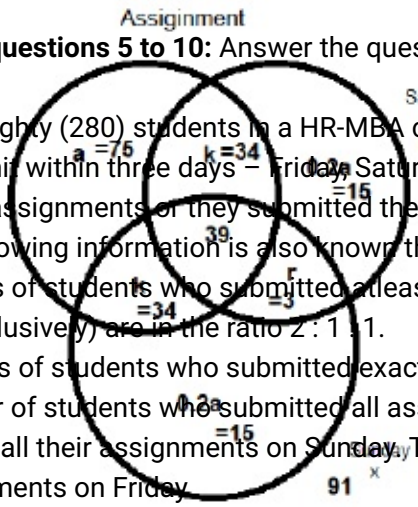
Step 4:

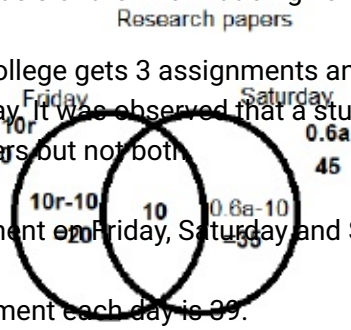
$1.4a + 2k + r + 39 + 10r - 10 + 10 + 0.6a - 10 = 280$

⇒ 2a + 2k + 11r = 251 ... (1)
a + 2k + 39 = 2(k + r + 39 + 0.2a) ... (2)
⇒ 0.6a – 2r = 39 + 2r ... (2)
0.6a + 2(0.6a – 10) × 10 = 1300
⇒ 150r + 6a = 900 ... (3)
From (2) and (3), we get
r = 3, a = 75
Putting these values in (1), we get
2a + 2k + 11r = 251
⇒ k = 34
Hence, final venn diagrams will look like:

The grace marks awarded by the professor in the research papers submitted on Saturday is exactly = (35 × 2 + 10) × 10 = 800.
Hence, it cannot be more than 800.

Bookmark FeedBack





Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Two hundred eighty (280) students in a HR-MBA class of XYZ college gets 3 assignments and 2 research papers to submit within three days – Friday, Saturday and Sunday. It was observed that a student either submitted the assignments or they submitted the research papers but not both.
Further, the following information is also known that:
(i) The numbers of students who submitted atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.
(ii) The numbers of students who submitted exactly one assignment each day is 39.
(iii) The number of students who submitted all assignments on Saturday is the same as the number of students who submitted all their assignments on Sunday. This number is 20% of the number of students who submitted all their assignments on Friday.
(iv) No student submitted any research paper on Sunday. Of all the students who submitted the research papers, only 10 students summitted the research papers on both days.
(v) The number of students who submitted atleast one research paper on Saturday is 60% of the students who submitted all the assignments on Friday.
(vi) The number of students who submitted at least one research paper on Friday and the number of students who submitted all 3 assignments in two days Saturday and Sunday only are in the ratio of 10 : 1.

The professor decides to award grace marks per submission to the students based on how early they submit the assignment or research paper.

	Friday	Saturday	Sunday
Grace marks per Assignment	10	5	1
Grace marks per Research paper	15	10	

The total grace marks professor awarded to the students who submitted all research papers either on Friday or Saturday is 1300.

Q.10 [11831809]
What is the difference between the number of students who submitted the assignments and the number of students who submitted the research papers?

1 ◯ 150

2 ◯ 65

3 ◯ 215

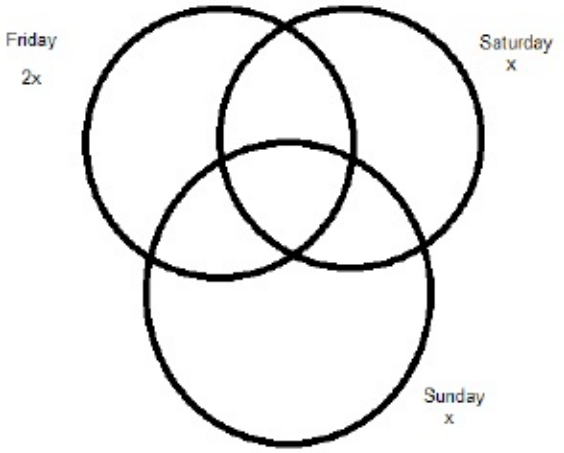
4 ◯ 130

Solution:
Correct Answer : 1

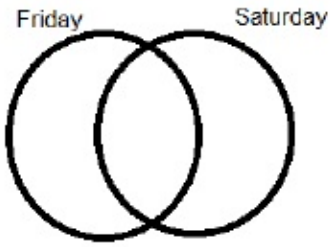
Answer key/Solution

Step 1:
From condition (i), The numbers of students who submitted the atleast one assignment on Friday, Saturday and Sunday (though may be not exclusively) are in the ratio 2 : 1 : 1.

Assiginment

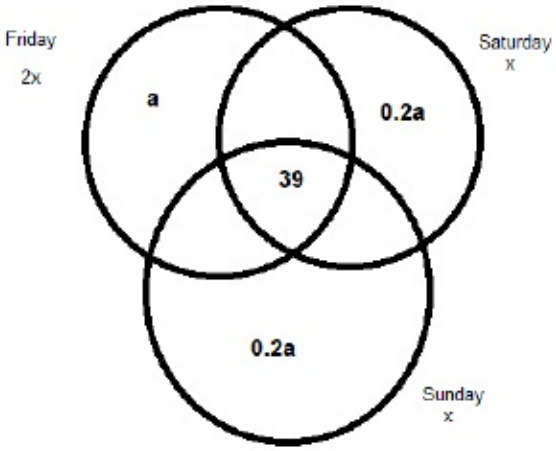


Research papers

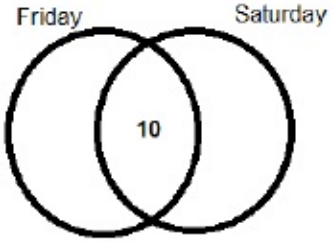


Step 2:
From conditions (ii) and (iii), The numbers of students who submitted assignment exactly one assignment every day is 39. Let number of students submitting all three assignments on Friday be 'a'.

Assiginment

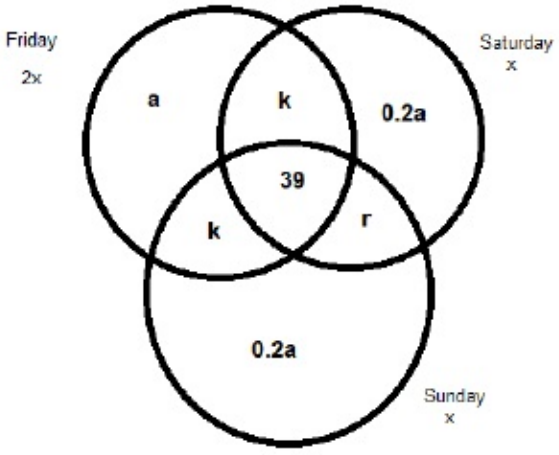


Research papers

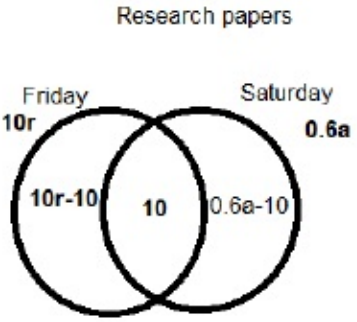


Step 3:
Also, from condition (iv), (v) and (vi), we get

Assiginment



Research papers



Number of students who submitted atleast one assignment on Saturday is equal to number of students who submitted exactly one assignment on Sunday. Therefore,

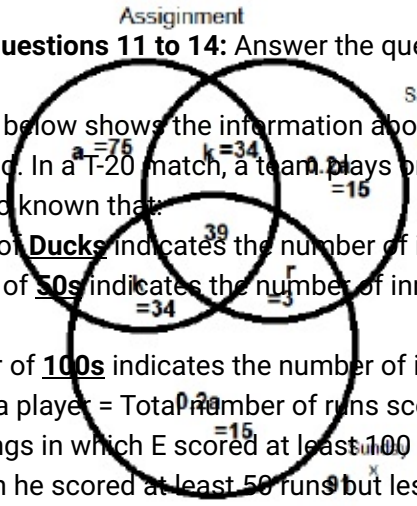
$M + 0.2a + 39 + r = N + 39 + r + 0.2a$

$\Rightarrow M = N = k.$

Step 4:

$1.4a + 2k + r + 39 + 10r - 10 + 10 + 0.6a - 10 = 280$

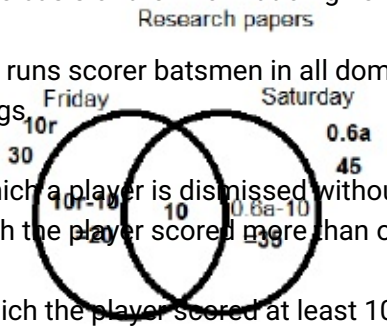
⇒ 2a + 2k + 11r = 251 ... (1)
a + 2k + 39 = 2(k + r + 39 + 0.2a) ... (2)
⇒ 0.6a = 39 + 2r
⇒ 150r + 6a = 900 ... (3)
From (2) and (3), we get
r = 3, a = 75
Putting these values in (1), we get
2a + 2k + 11r = 251
⇒ k = 34
Hence, final venn diagrams will look like:



The number of students who submitted the assignments = 215
The number of students who submitted the research papers = 65
Hence, required difference = 215 – 65 = 150.

Bookmark FeedBack

Directions for questions 11 to 14: Answer the questions on the basis of the information given below.



The table given below shows the information about the top 10 runs scorer batsmen in all domestic T-20 leagues around the world. In a T-20 match, a team plays only one innings.
Further, it is also known that:
(i) The number of **Ducks** indicates the number of innings in which a player is dismissed without scoring a run.
(ii) The number of **50s** indicates the number of innings in which the player scored more than or equal to 50 but less than 100.
(iii) The number of **100s** indicates the number of innings in which the player scored at least 100 runs.
(iv) Average of a player = Total number of runs scored by the player/Number of innings played by the player
(v) In each innings in which E scored at least 100 runs, each six that he hit covered exactly 110 meters. In each innings in which he scored at least 50 runs but less than 100 runs, each six that he hit covered exactly 105 meters and in each of the other innings, each six that he hit covered exactly 100 meters. Further, in each innings in which he scored at least 100 runs, he hit at most 10 sixes and in each innings in which he scored at least 50 runs but less than 100 runs, he hit at most 5 sixes.

Player	No. of Innings	Runs Scored	Balls faced	No. of 50s	No. of 100s	No. of 4's	No. of 6's	No. of Ducks
A	121	7437	5611	23	14	987	71	4
B	109	6512	4345	7	24	761	181	6
C	126	7657	5738	29	7	999	109	3
D	139	6677	3684	26	4	932	169	4
E	154	8710	4612	14	11	1013	200	9
F	102	5252	2919	22	7	707	139	5
G	105	4810	2894	16	4	688	144	13
H	120	6230	3497	17	8	735	149	10
I	120	6024	3989	12	12	841	128	6
J	159	4755	2112	21	2	419	199	2

Q.11 [11831809]
If in no innings, A hit more than ten 4's, then what could be the maximum number of innings in which he hit at least one 4 but not more than four 4s?

Solution:
Correct Answer : 30

Answer key/Solution

To maximize the numbers of innings with at most four 4s and at least one 4, maximize the number of innings in which he hit exactly ten 4s and the number of innings in which he hit exactly four 4s.
Let the number of innings in which he hit ten 4s be x, then the number of innings in which he hit exactly four 4s be (117 – x), since there are 4 innings in which he scored a Duck.
$\Rightarrow 10x + 4(117 - x) = 987$
$\Rightarrow x = 86.5$
Taking x = 87, the maximum number of innings in which he hit at least one 4 and at most four 4s is (117 – 87) i.e., 30.
(**Note:** 'x' cannot be a fraction; and if x is taken as 86 and (117 – x) as 31, then the number of fours that he hit could at maximum 984. Hence, 'x' will be taken as 87.)

Bookmark FeedBack

Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

The table given below shows the information about the top 10 runs scorer batsmen in all domestic T-20 leagues around the world. In a T-20 match, a team plays only one innings.
Further, it is also known that:
(i) The number of **Ducks** indicates the number of innings in which a player is dismissed without scoring a run.
(ii) The number of **50s** indicates the number of innings in which the player scored more than or equal to 50 but less than 100.
(iii) The number of **100s** indicates the number of innings in which the player scored at least 100 runs.
(iv) Average of a player = Total number of runs scored by the player/Number of innings played by the player
(v) In each innings in which E scored at least 100 runs, each six that he hit covered exactly 110 meters. In each innings in which he scored at least 50 runs but less than 100 runs, each six that he hit covered exactly 105 meters and in each of the other innings, each six that he hit covered exactly 100 meters. Further, in each innings in which he scored at least 100 runs, he hit at most 10 sixes and in each innings in which he scored at least 50 runs but less than 100 runs, he hit at most 5 sixes.

Player	No. of Innings	Runs Scored	Balls faced	No. of 50s	No. of 100s	No. of 4's	No. of 6's	No. of Ducks
A	121	7437	5611	23	14	987	71	4
B	109	6512	4345	7	24	761	181	6
C	126	7657	5738	29	7	999	109	3
D	139	6677	3684	26	4	932	169	4
E	154	8710	4612	14	11	1013	200	9
F	102	5252	2919	22	7	707	139	5
G	105	4810	2894	16	4	688	144	13
H	120	6230	3497	17	8	735	149	10
I	120	6024	3989	12	12	841	128	6
J	159	4755	2112	21	2	419	199	2

Q.12 [11831809]

Which player has the second highest strike rate among all the given players? (Strike rate is the runs scored per 100 balls faced.)

1 ◯ D

2 ◯ E

3 ◯ F

4 ◯ H

Solution:

Correct Answer : 2

Answer key/Solution

Checking with options,
For D, it is = 6677/3684 × 100 ≈ 181.24
For E, it is = 8710/4612 × 100 ≈ 188.85
For F, it is = 5252/2919 × 100 ≈ 179.92
For H, it is = 6230/3497 × 100 ≈ 178.15
For J, it is = 4755/2112 × 100 ≈ 225.14
And it is very clear, that J has the highest strike rate.
Hence, the 2nd highest strike rate is of E.

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Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

The table given below shows the information about the top 10 runs scorer batsmen in all domestic T-20 leagues around the world. In a T-20 match, a team plays only one innings.
Further, it is also known that:
(i) The number of **Ducks** indicates the number of innings in which a player is dismissed without scoring a run.
(ii) The number of **50s** indicates the number of innings in which the player scored more than or equal to 50 but less than 100.
(iii) The number of **100s** indicates the number of innings in which the player scored at least 100 runs.
(iv) Average of a player = Total number of runs scored by the player/Number of innings played by the player
(v) In each innings in which E scored at least 100 runs, each six that he hit covered exactly 110 meters. In each innings in which he scored at least 50 runs but less than 100 runs, each six that he hit covered exactly 105 meters and in each of the other innings, each six that he hit covered exactly 100 meters. Further, in each innings in which he scored at least 100 runs, he hit at most 10 sixes and in each innings in which he scored at least 50 runs but less than 100 runs, he hit at most 5 sixes.

Player	No. of Innings	Runs Scored	Balls faced	No. of 50s	No. of 100s	No. of 4's	No. of 6's	No. of Ducks
A	121	7437	5611	23	14	987	71	4
B	109	6512	4345	7	24	761	181	6
C	126	7657	5738	29	7	999	109	3
D	139	6677	3684	26	4	932	169	4
E	154	8710	4612	14	11	1013	200	9
F	102	5252	2919	22	7	707	139	5
G	105	4810	2894	16	4	688	144	13
H	120	6230	3497	17	8	735	149	10
I	120	6024	3989	12	12	841	128	6
J	159	4755	2112	21	2	419	199	2

Q.13 [11831809]
What could be the maximum approximate average of B in the innings in which he neither scored a 50 nor scored a 100?

1 ◯ 43.65

2 ◯ 49.12

3 ◯ 48.23

4 ◯ 47.20

Solution:
Correct Answer : 3

Answer key/Solution

To maximize his average, for innings in which he neither scored a 50 nor scored a hundred, take each innings in which he scored a 50 to be a exactly 50 runs (not more) and each innings in which he scored a 100 to be of exactly 100 runs (not more).
There are 7 innings in which he scored a 50 and 24 innings in which he scored a 100, which equates to a minimum of = [(7 × 50) + (24 × 100)] runs;
Remaining runs = 6512 – (350 + 2400) = 3762
Remaining innings = 109 – (24 + 7) = 78
Hence, his maximum average = 3762/78 ≈ 48.23.

Bookmark FeedBack

Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

The table given below shows the information about the top 10 runs scorer batsmen in all domestic T-20 leagues around the world. In a T-20 match, a team plays only one innings.
Further, it is also known that:
(i) The number of **Ducks** indicates the number of innings in which a player is dismissed without scoring a run.
(ii) The number of **50s** indicates the number of innings in which the player scored more than or equal to 50 but less than 100.
(iii) The number of **100s** indicates the number of innings in which the player scored at least 100 runs.
(iv) Average of a player = Total number of runs scored by the player/Number of innings played by the player
(v) In each innings in which E scored at least 100 runs, each six that he hit covered exactly 110 meters. In each innings in which he scored at least 50 runs but less than 100 runs, each six that he hit covered exactly 105 meters and in each of the other innings, each six that he hit covered exactly 100 meters. Further, in each innings in which he scored at least 100 runs, he hit at most 10 sixes and in each innings in which he scored at least 50 runs but less than 100 runs, he hit at most 5 sixes.

Player	No. of Innings	Runs Scored	Balls faced	No. of 50s	No. of 100s	No. of 4's	No. of 6's	No. of Ducks
A	121	7437	5611	23	14	987	71	4
B	109	6512	4345	7	24	761	181	6
C	126	7657	5738	29	7	999	109	3
D	139	6677	3684	26	4	932	169	4
E	154	8710	4612	14	11	1013	200	9
F	102	5252	2919	22	7	707	139	5
G	105	4810	2894	16	4	688	144	13
H	120	6230	3497	17	8	735	149	10
I	120	6024	3989	12	12	841	128	6
J	159	4755	2112	21	2	419	199	2

Q.14 [11831809]

What could be the maximum value of the total distance (in meters) covered by all the sixes that E hit?

1 ◯ 19,865

2 ◯ 20,340

3 ◯ 18,380

4 ◯ 21,450

Solution:

Correct Answer : 4

Answer key/Solution

To maximize the total distances covered by all the sixes that E hit, maximize the number of sixes in each innings in which he scored at least 100 runs (which is 10 sixes per such innings) and also maximize the number of sixes in each innings in which he scored at least 50 runs but less than 100 runs (which is 5 sixes per such innings). Hence, the maximum distance covered is = 11 × 10 × 110 + 14 × 5 × 105 + 20 × 100
= 12100 + 7350 + 2000 = 21450 meters.

Bookmark FeedBack

Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

At a selection camp for Jawans in Multifunctional Armed Forces there are three sections of tests - Ground, Aerial and Underwater . The test is open for all types of candidates - Civilian, Army, Airforce and Navy. A total of 15 Jawans - J01, J02, …, J15 - appear for the tests. The maximum score for each test is 10 and all scores are given as whole numbers. Any score below 5 calls for rejection.

A Civilian first appears for the Ground test and then for the Aerial test. If he passes in both tests, then he is selected and he need not appear for the Underwater test. If he fails in both, then he is rejected. If the Civilian fails in one and passes in the other, then he appears for the Underwater test and has to pass to be selected.

An Army jawan appears for the Aerial test first. If cleared, then he is selected. If not, then he has to clear the Underwater test to qualify.

An Airforce jawan appears for the Underwater test first. If cleared, then he is selected. If not, then he has to clear the Ground test to qualify.

A Navy jawan appears for the Ground test first. If cleared, then he is selected. If not, then he has to clear the Aerial test to qualify.

The following facts are also known about the selection process after its completion.
(i) J02, J06, J09, J11, and J12 were rejected and all others were selected.
(ii) Ground tests were taken by only J02, J03, J04, J06, J08, J10, J11, J13 and J14.
(iii) Aerial tests were taken by only J02, J04, J06 through J09, J12 and J13.
(iv) Underwater tests were taken by only J01 through J05, J07, J08, J09, J11, J12 and J15.
(v) J01, J05, J07 and J08 got a score of 9 out of 10 in the same test.
(vi) J10, J13, J14 and J15 got maximum possible scores in one of the tests.

Q.15 [11831809]
For which of the following pairs of jawans can a total score of more than 15 be possible?

1 ◯ J08 - J13

2 ◯ J04 - J08

3 ◯ J14 - J13

4 ◯ J04 - J15

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Based on the information given, we can say that if a Civilian candidate appears for the test, then six cases are possible:

Ground	Aerial	Underwater	Selected / Rejected
✓	✓	-	✓
✓	✗	✓	✓
✓	✗	✗	✗
✗	✓	✓	✓
✗	✓	✗	✗
✗	✗	-	✗

If an Army jawan appears, he takes the Aerial test first and if he fails, he takes the Underwater test.
If an Airforce jawan appears, he takes the Underwater test first and if he fails, he takes the Ground test.
If a Navy jawan appears, he takes the Ground test first and if he fails, he takes the Aerial test.
Note that there is a cyclic pattern in the tests taken..

Step 2:
Based on the information given regarding the questions, we can create the following table:

Jawan	Selected or Rejected	Ground test	Aerial test	Underwater test	Total score	Category
J01	✓	-	-	✓ 9	9	Airforce
J02	✗	✗/✓	✗/✓	✗	-	Civilian
J03	✓	✓	-	✗	-	Airforce
J04	✓	✗/✓	✗/✓	✓		Civilian
J05	✓	-	-	✓ 9	9	Airforce
J06	✗	✗	✗	-		Civilian / Navy
J07	✓	-	✗	✓ 9	9	Army
J08	✓	✗/✓	✗/✓	✓ 9		Civilian
J09	✗	-	✗	✗	-	Army
J10	✓	✓ 10	-	-	10	Navy
J11	✗	✗	-	✗	-	Airforce
J12	✗	-	✗	✗	-	Army
J13	✓	✗	✓ 10	-	10	Navy
J14	✓	✓ 10	-	-	10	Navy
J15	✓	-	-	✓ 10	10	Airforce

Jawans J04 and J08 are Civilians who got selected, so they definitely appear for 2 or more tests and get scores more than 5 in each. So it is possible for them to get a total score of more than 15.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

At a selection camp for Jawans in Multifunctional Armed Forces there are three sections of tests - Ground, Aerial and Underwater . The test is open for all types of candidates - Civilian, Army, Airforce and Navy. A total of 15 Jawans - J01, J02, …, J15 - appear for the tests. The maximum score for each test is 10 and all scores are given as whole numbers. Any score below 5 calls for rejection.

A Civilian first appears for the Ground test and then for the Aerial test. If he passes in both tests, then he is selected and he need not appear for the Underwater test. If he fails in both, then he is rejected. If the Civilian fails in one and passes in the other, then he appears for the Underwater test and has to pass to be selected.

An Army jawan appears for the Aerial test first. If cleared, then he is selected. If not, then he has to clear the Underwater test to qualify.

An Airforce jawan appears for the Underwater test first. If cleared, then he is selected. If not, then he has to clear the Ground test to qualify.

A Navy jawan appears for the Ground test first. If cleared, then he is selected. If not, then he has to clear the Aerial test to qualify.

The following facts are also known about the selection process after its completion.
(i) J02, J06, J09, J11, and J12 were rejected and all others were selected.
(ii) Ground tests were taken by only J02, J03, J04, J06, J08, J10, J11, J13 and J14.
(iii) Aerial tests were taken by only J02, J04, J06 through J09, J12 and J13.
(iv) Underwater tests were taken by only J01 through J05, J07, J08, J09, J11, J12 and J15.
(v) J01, J05, J07 and J08 got a score of 9 out of 10 in the same test.
(vi) J10, J13, J14 and J15 got maximum possible scores in one of the tests.

Q.16 [11831809]
How many jawans were from the Airforce?

Solution:
Correct Answer : 5

Answer key/Solution

Step 1:
Based on the information given, we can say that if a Civilian candidate appears for the test, then six cases are possible:

Ground	Aerial	Underwater	Selected / Rejected
✓	✓	-	✓
✓	✗	✓	✓
✓	✗	✗	✗
✗	✓	✓	✓
✗	✓	✗	✗
✗	✗	-	✗

If an Army jawan appears, he takes the Aerial test first and if he fails, he takes the Underwater test.
If an Airforce jawan appears, he takes the Underwater test first and if he fails, he takes the Ground test.
If a Navy jawan appears, he takes the Ground test first and if he fails, he takes the Aerial test.
Note that there is a cyclic pattern in the tests taken..

Step 2:
Based on the information given regarding the questions, we can create the following table:

Jawan	Selected or Rejected	Ground test	Aerial test	Underwater test	Total score	Category
J01	✓	-	-	✓ 9	9	Airforce
J02	✗	✗/✓	✗/✓	✗	-	Civilian
J03	✓	✓	-	✗	-	Airforce
J04	✓	✗/✓	✗/✓	✓		Civilian
J05	✓	-	-	✓ 9	9	Airforce
J06	✗	✗	✗	-		Civilian / Navy
J07	✓	-	✗	✓ 9	9	Army
J08	✓	✗/✓	✗/✓	✓ 9		Civilian
J09	✗	-	✗	✗	-	Army
J10	✓	✓ 10	-	-	10	Navy
J11	✗	✗	-	✗	-	Airforce
J12	✗	-	✗	✗	-	Army
J13	✓	✗	✓ 10	-	10	Navy
J14	✓	✓ 10	-	-	10	Navy
J15	✓	-	-	✓ 10	10	Airforce

From the table we can say that five jawans were from the Airforce.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

At a selection camp for Jawans in Multifunctional Armed Forces there are three sections of tests - Ground, Aerial and Underwater . The test is open for all types of candidates - Civilian, Army, Airforce and Navy. A total of 15 Jawans - J01, J02, …, J15 - appear for the tests. The maximum score for each test is 10 and all scores are given as whole numbers. Any score below 5 calls for rejection.

A Civilian first appears for the Ground test and then for the Aerial test. If he passes in both tests, then he is selected and he need not appear for the Underwater test. If he fails in both, then he is rejected. If the Civilian fails in one and passes in the other, then he appears for the Underwater test and has to pass to be selected.

An Army jawan appears for the Aerial test first. If cleared, then he is selected. If not, then he has to clear the Underwater test to qualify.

An Airforce jawan appears for the Underwater test first. If cleared, then he is selected. If not, then he has to clear the Ground test to qualify.

A Navy jawan appears for the Ground test first. If cleared, then he is selected. If not, then he has to clear the Aerial test to qualify.

The following facts are also known about the selection process after its completion.
(i) J02, J06, J09, J11, and J12 were rejected and all others were selected.
(ii) Ground tests were taken by only J02, J03, J04, J06, J08, J10, J11, J13 and J14.
(iii) Aerial tests were taken by only J02, J04, J06 through J09, J12 and J13.
(iv) Underwater tests were taken by only J01 through J05, J07, J08, J09, J11, J12 and J15.
(v) J01, J05, J07 and J08 got a score of 9 out of 10 in the same test.
(vi) J10, J13, J14 and J15 got maximum possible scores in one of the tests.

Q.17 [11831809]
What is the difference between the maximum and the minimum scores of any Civilian who is selected?

1 ◯ 16

2 ◯ 8

3 ◯ 10

4 ◯ 12

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Based on the information given, we can say that if a Civilian candidate appears for the test, then six cases are possible:

Ground	Aerial	Underwater	Selected / Rejected
✓	✓	-	✓
✓	×	✓	✓
✓	×	×	×
×	✓	✓	✓
×	✓	×	×
×	×	-	×

If an Army jawan appears, he takes the Aerial test first and if he fails, he takes the Underwater test.
If an Airforce jawan appears, he takes the Underwater test first and if he fails, he takes the Ground test.
If a Navy jawan appears, he takes the Ground test first and if he fails, he takes the Aerial test.
Note that there is a cyclic pattern in the tests taken..

Step 2:
Based on the information given regarding the questions, we can create the following table:

Jawan	Selected or Rejected	Ground test	Aerial test	Underwater test	Total score	Category
J01	✓	-	-	✓ 9	9	Airforce
J02	×	×/✓	×/✓	×	-	Civilian
J03	✓	✓	-	×	-	Airforce
J04	✓	×/✓	×/✓	✓		Civilian
J05	✓	-	-	✓ 9	9	Airforce
J06	×	×	×	-		Civilian / Navy
J07	✓	-	×	✓ 9	9	Army
J08	✓	×/✓	×/✓	✓ 9		Civilian
J09	×	-	×	×	-	Army
J10	✓	✓ 10	-	-	10	Navy
J11	×	×	-	×	-	Airforce
J12	×	-	×	×	-	Army
J13	✓	×	✓ 10	-	10	Navy
J14	✓	✓ 10	-	-	10	Navy
J15	✓	-	-	✓ 10	10	Airforce

Let us take J04, a Civilian, who took all three tests and was selected.
For the maximum score we take a perfect 10 in the qualifying tests and a score of 4 in the one that he failed. So his maximum total score can be 24.
For the minimum score we take a score of 5 in the qualifying tests and 0 in the one that he failed. So the minimum total score can be 10.
Let us take J08, a Civilian, who took all three tests and was selected.
The maximum score = 10 + 9 + 4 = 23 and minimum score = 5 + 9 + 0 = 14
Hence, required difference = 24 – 14 = 10.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

At a selection camp for Jawans in Multifunctional Armed Forces there are three sections of tests - Ground, Aerial and Underwater . The test is open for all types of candidates - Civilian, Army, Airforce and Navy. A total of 15 Jawans - J01, J02, ..., J15 - appear for the tests. The maximum score for each test is 10 and all scores are given as whole numbers. Any score below 5 calls for rejection.

A Civilian first appears for the Ground test and then for the Aerial test. If he passes in both tests, then he is selected and he need not appear for the Underwater test. If he fails in both, then he is rejected. If the Civilian fails in one and passes in the other, then he appears for the Underwater test and has to pass to be selected.

An Army jawan appears for the Aerial test first. If cleared, then he is selected. If not, then he has to clear the Underwater test to qualify.

An Airforce jawan appears for the Underwater test first. If cleared, then he is selected. If not, then he has to clear the Ground test to qualify.

A Navy jawan appears for the Ground test first. If cleared, then he is selected. If not, then he has to clear the Aerial test to qualify.

The following facts are also known about the selection process after its completion.
(i) J02, J06, J09, J11, and J12 were rejected and all others were selected.
(ii) Ground tests were taken by only J02, J03, J04, J06, J08, J10, J11, J13 and J14.
(iii) Aerial tests were taken by only J02, J04, J06 through J09, J12 and J13.
(iv) Underwater tests were taken by only J01 through J05, J07, J08, J09, J11, J12 and J15.
(v) J01, J05, J07 and J08 got a score of 9 out of 10 in the same test.
(vi) J10, J13, J14 and J15 got maximum possible scores in one of the tests.

Q.18 [11831809]
How many jawans DEFINITELY failed in the Aerial test?

1 ◯ 3

2 ◯ 4

3 ◯ 5

4 ◯ 7

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Based on the information given, we can say that if a Civilian candidate appears for the test, then six cases are possible:

Ground	Aerial	Underwater	Selected / Rejected
✓	✓	-	✓
✓	✗	✓	✓
✓	✗	✗	✗
✗	✓	✓	✓
✗	✓	✗	✗
✗	✗	-	✗

If an Army jawan appears, he takes the Aerial test first and if he fails, he takes the Underwater test.
If an Airforce jawan appears, he takes the Underwater test first and if he fails, he takes the Ground test.
If a Navy jawan appears, he takes the Ground test first and if he fails, he takes the Aerial test.
Note that there is a cyclic pattern in the tests taken..

Step 2:
Based on the information given regarding the questions, we can create the following table:

Jawan	Selected or Rejected	Ground test	Aerial test	Underwater test	Total score	Category
J01	✓	-	-	✓ 9	9	Airforce
J02	✗	✗/✓	✗/✓	✗	-	Civilian
J03	✓	✓	-	✗	-	Airforce
J04	✓	✗/✓	✗/✓	✓		Civilian
J05	✓	-	-	✓ 9	9	Airforce
J06	✗	✗	✗	-		Civilian / Navy
J07	✓	-	✗	✓ 9	9	Army
J08	✓	✗/✓	✗/✓	✓ 9		Civilian
J09	✗	-	✗	✗	-	Army
J10	✓	✓ 10	-	-	10	Navy
J11	✗	✗	-	✗	-	Airforce
J12	✗	-	✗	✗	-	Army
J13	✓	✗	✓ 10	-	10	Navy
J14	✓	✓ 10	-	-	10	Navy
J15	✓	-	-	✓ 10	10	Airforce

Four jawans (J06, J07, J09, J12) definitely failed in the Aerial test.

Bookmark FeedBack

Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

At a selection camp for Jawans in Multifunctional Armed Forces there are three sections of tests - Ground, Aerial and Underwater . The test is open for all types of candidates - Civilian, Army, Airforce and Navy. A total of 15 Jawans - J01, J02, …, J15 - appear for the tests. The maximum score for each test is 10 and all scores are given as whole numbers. Any score below 5 calls for rejection.

A Civilian first appears for the Ground test and then for the Aerial test. If he passes in both tests, then he is selected and he need not appear for the Underwater test. If he fails in both, then he is rejected. If the Civilian fails in one and passes in the other, then he appears for the Underwater test and has to pass to be selected.

An Army jawan appears for the Aerial test first. If cleared, then he is selected. If not, then he has to clear the Underwater test to qualify.

An Airforce jawan appears for the Underwater test first. If cleared, then he is selected. If not, then he has to clear the Ground test to qualify.

A Navy jawan appears for the Ground test first. If cleared, then he is selected. If not, then he has to clear the Aerial test to qualify.

The following facts are also known about the selection process after its completion.
(i) J02, J06, J09, J11, and J12 were rejected and all others were selected.
(ii) Ground tests were taken by only J02, J03, J04, J06, J08, J10, J11, J13 and J14.
(iii) Aerial tests were taken by only J02, J04, J06 through J09, J12 and J13.
(iv) Underwater tests were taken by only J01 through J05, J07, J08, J09, J11, J12 and J15.
(v) J01, J05, J07 and J08 got a score of 9 out of 10 in the same test.
(vi) J10, J13, J14 and J15 got maximum possible scores in one of the tests.

Q.19 [11831809]
The success ratio of a test is the ratio of the number of jawans that passed the test to the number of jawans that appeared for the test. Which option best describes the success ratio of the Ground test?

1 ◯ lies between 0.66 and 0.99

2 ◯ either 0.33 or 0.66

3 ◯ 0.33

4 ◯ lies between 0.33 and 0.66 (both inclusive)

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Based on the information given, we can say that if a Civilian candidate appears for the test, then six cases are possible:

Ground	Aerial	Underwater	Selected / Rejected
✓	✓	-	✓
✓	✗	✓	✓
✓	✗	✗	✗
✗	✓	✓	✓
✗	✓	✗	✗
✗	✗	-	✗

If an Army jawan appears, he takes the Aerial test first and if he fails, he takes the Underwater test.
If an Airforce jawan appears, he takes the Underwater test first and if he fails, he takes the Ground test.
If a Navy jawan appears, he takes the Ground test first and if he fails, he takes the Aerial test.
Note that there is a cyclic pattern in the tests taken..

Step 2:
Based on the information given regarding the questions, we can create the following table:

Jawan	Selected or Rejected	Ground test	Aerial test	Underwater test	Total score	Category
J01	✓	-	-	✓ 9	9	Airforce
J02	✗	✗/✓	✗/✓	✗	-	Civilian
J03	✓	✓	-	✗	-	Airforce
J04	✓	✗/✓	✗/✓	✓		Civilian
J05	✓	-	-	✓ 9	9	Airforce
J06	✗	✗	✗	-		Civilian / Navy
J07	✓	-	✗	✓ 9	9	Army
J08	✓	✗/✓	✗/✓	✓ 9		Civilian
J09	✗	-	✗	✗	-	Army
J10	✓	✓ 10	-	-	10	Navy
J11	✗	✗	-	✗	-	Airforce
J12	✗	-	✗	✗	-	Army
J13	✓	✗	✓ 10	-	10	Navy
J14	✓	✓ 10	-	-	10	Navy
J15	✓	-	-	✓ 10	10	Airforce

The minimum success ratio of the Ground test = 3/9 = 0.33;
The maximum success ratio of the Ground test = 6/9 = 0.66.
Hence, it lies between 0.33 and 0.66 (both inclusive).

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

At a selection camp for Jawans in Multifunctional Armed Forces there are three sections of tests - Ground, Aerial and Underwater . The test is open for all types of candidates - Civilian, Army, Airforce and Navy. A total of 15 Jawans - J01, J02, …, J15 - appear for the tests. The maximum score for each test is 10 and all scores are given as whole numbers. Any score below 5 calls for rejection.

A Civilian first appears for the Ground test and then for the Aerial test. If he passes in both tests, then he is selected and he need not appear for the Underwater test. If he fails in both, then he is rejected. If the Civilian fails in one and passes in the other, then he appears for the Underwater test and has to pass to be selected.

An Army jawan appears for the Aerial test first. If cleared, then he is selected. If not, then he has to clear the Underwater test to qualify.

An Airforce jawan appears for the Underwater test first. If cleared, then he is selected. If not, then he has to clear the Ground test to qualify.

A Navy jawan appears for the Ground test first. If cleared, then he is selected. If not, then he has to clear the Aerial test to qualify.

The following facts are also known about the selection process after its completion.
(i) J02, J06, J09, J11, and J12 were rejected and all others were selected.
(ii) Ground tests were taken by only J02, J03, J04, J06, J08, J10, J11, J13 and J14.
(iii) Aerial tests were taken by only J02, J04, J06 through J09, J12 and J13.
(iv) Underwater tests were taken by only J01 through J05, J07, J08, J09, J11, J12 and J15.
(v) J01, J05, J07 and J08 got a score of 9 out of 10 in the same test.
(vi) J10, J13, J14 and J15 got maximum possible scores in one of the tests.

Q.20 [11831809]
How many jawans from the Navy did not fail in any of the tests that they appeared for?

1 ◯ 2

2 ◯ 5

3 ◯ 4

4 ◯ 3

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Based on the information given, we can say that if a Civilian candidate appears for the test, then six cases are possible:

Ground	Aerial	Underwater	Selected / Rejected
✓	✓	-	✓
✓	✗	✓	✓
✓	✗	✗	✗
✗	✓	✓	✓
✗	✓	✗	✗
✗	✗	-	✗

If an Army jawan appears, he takes the Aerial test first and if he fails, he takes the Underwater test.
If an Airforce jawan appears, he takes the Underwater test first and if he fails, he takes the Ground test.
If a Navy jawan appears, he takes the Ground test first and if he fails, he takes the Aerial test.
Note that there is a cyclic pattern in the tests taken..

Step 2:
Based on the information given regarding the questions, we can create the following table:

Jawan	Selected or Rejected	Ground test	Aerial test	Underwater test	Total score	Category
J01	✓	-	-	✓ 9	9	Airforce
J02	✗	✗/✓	✗/✓	✗	-	Civilian
J03	✓	✓	-	✗	-	Airforce
J04	✓	✗/✓	✗/✓	✓		Civilian
J05	✓	-	-	✓ 9	9	Airforce
J06	✗	✗	✗	-		Civilian / Navy
J07	✓	-	✗	✓ 9	9	Army
J08	✓	✗/✓	✗/✓	✓ 9		Civilian
J09	✗	-	✗	✗	-	Army
J10	✓	✓ 10	-	-	10	Navy
J11	✗	✗	-	✗	-	Airforce
J12	✗	-	✗	✗	-	Army
J13	✓	✗	✓ 10	-	10	Navy
J14	✓	✓ 10	-	-	10	Navy
J15	✓	-	-	✓ 10	10	Airforce

It is clear from the table that J10 and J14 are two jawans from the Navy who did not fail in any test they appeared for.

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